

DO NOT BEGIN THIS EXAM UNTIL TOLD TO START

Name: _____ **Solution** _____

PeopleSoft ID: _____

ECE 3317
Applied Electromagnetic Waves

Exam II

Nov. 28, 2018

1. This exam is open book and open notes. However, you are not allowed to use a computer or any electronic device other than a calculator. Any devices that may be used to communicate are not allowed.
2. Show all of your work. No credit will be given if the work required to obtain the solutions is not shown.
3. Perform all your work on the exam in the space allowed.
4. Write neatly. You will not be given credit for work that is not **easily legible**.
5. Leave answers in terms of the parameters given in the problem.
6. Show units in all of your final answers.
7. Circle your final answers.
8. Double-check your answers. For simpler problems, partial credit may not be given.
9. If you have any questions, ask the instructor. You will not be given credit for work that is based on a wrong assumption.
10. Make sure you sign the academic honesty statement on the next page.

Academic Honesty Statement

I agree to abide by the UH Academic Honesty Policy during this exam. I understand that the punishment for violating this policy will be most severe, including getting an F in the class and getting expelled from the University.

Signature

FORMULA SHEET

$$\nabla \times \underline{\mathcal{E}} = -\frac{\partial \underline{\mathcal{B}}}{\partial t}$$

$$\nabla \times \underline{\mathcal{H}} = \underline{\mathcal{J}} + \frac{\partial \underline{\mathcal{D}}}{\partial t}$$

$$\nabla \cdot \underline{\mathcal{B}} = 0$$

$$\nabla \cdot \underline{\mathcal{D}} = \rho_v$$

$$\nabla \times \underline{E} = -j\omega \underline{B}$$

$$\nabla \times \underline{H} = \underline{J} + j\omega \underline{D}$$

$$\nabla \cdot \underline{B} = 0$$

$$\nabla \cdot \underline{D} = \rho_v$$

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

$$\epsilon_0 \doteq 8.8541878 \times 10^{-12} \text{ [F/m]}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ [H/m]}$$

$$\oint_S (\underline{\mathcal{E}} \times \underline{\mathcal{H}}) \cdot \hat{n} dS = -\int_V \sigma |\underline{\mathcal{E}}|^2 dV - \int_V \frac{\partial}{\partial t} \left(\frac{1}{2} \mu |\underline{\mathcal{H}}|^2 \right) dV - \int_V \frac{\partial}{\partial t} \left(\frac{1}{2} \epsilon |\underline{\mathcal{E}}|^2 \right) dV$$

$$\underline{\mathcal{S}} = \underline{\mathcal{E}} \times \underline{\mathcal{H}}$$

$$\underline{S} \equiv \frac{1}{2} (\underline{E} \times \underline{H}^*)$$

$$C = \frac{2\pi\epsilon_0\epsilon_r}{\ln\left(\frac{b}{a}\right)} \quad [\text{F/m}]$$

$$L = \frac{\mu_0}{2\pi} \ln\left(\frac{b}{a}\right) \quad [\text{H/m}]$$

$$G = \frac{2\pi\sigma_d}{\ln\left(\frac{b}{a}\right)} \quad [\text{S/m}]$$

$$R = \frac{1}{\sigma_m\delta} \left(\frac{1}{2\pi a} + \frac{1}{2\pi b} \right) \quad [\Omega/\text{m}]$$

$$\frac{\partial v}{\partial z} = -Ri - L \frac{\partial i}{\partial t}$$

$$\frac{\partial I}{\partial z} i = -Gv - C \frac{\partial v}{\partial t}$$

$$v(z,t) = f(z - c_d t) + g(z + c_d t)$$

$$i(z,t) = \frac{1}{Z_0} [f(z - c_d t) - g(z + c_d t)]$$

$$v(z,t) = v_g(t - z / c_d)$$

$$\Gamma_g = \left(\frac{R_g - Z_0}{R_g + Z_0} \right) \quad \Gamma_L = \left(\frac{R_L - Z_0}{R_L + Z_0} \right)$$

$$V^+ = \left(\frac{Z_0}{R_g + Z_0} \right) V_0$$

$$\Gamma_L(t) = 1 - 2e^{-(t-T)/\tau}, \quad t \geq T \quad \tau = Z_0 C_L$$

$$\Gamma_L(t) = -1 + 2e^{-(t-T)/\tau}, \quad t \geq T \qquad \tau = L_L / Z_0$$

$$V(z) = Ae^{-\gamma z} + Be^{+\gamma z}$$

$$\gamma = \sqrt{(R + j\omega L)(G + j\omega C)}$$

$$LC = \mu \varepsilon = \frac{1}{c_d^2}$$

$$\gamma = \alpha + j\beta$$

$$k_z = -j\gamma = \beta - j\alpha$$

$$v_p = \frac{\omega}{\beta}$$

$$\beta = \frac{2\pi}{\lambda}$$

$$\text{attenuation} = \left(\frac{20}{\ln 10}\right)\alpha = (8.6859)\alpha \quad [\text{dB/m}]$$

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

$$i(z) = \left(\frac{1}{Z_0}\right) \big[Ae^{-\gamma z} - Be^{+\gamma z} \big]$$

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$V(z) = A\Big(e^{-\gamma z} + \Gamma_L e^{+\gamma z}\Big)$$

$$I(z) = \frac{1}{Z_0} A \left(e^{-\gamma z} - \Gamma_L e^{+\gamma z} \right)$$

$$Z_{in}(z) = \frac{V(z)}{I(z)}$$

$$Z_{in}(z) = Z_0 \left(\frac{1 + \Gamma_L e^{+2\gamma z}}{1 - \Gamma_L e^{+2\gamma z}} \right)$$

$$Z_{in} = Z_0 \left(\frac{Z_L + Z_0 \tanh(\gamma l)}{Z_0 + Z_L \tanh(\gamma l)} \right)$$

$$Z_{in} = Z_0 \left(\frac{Z_L + jZ_0 \tan(\beta l)}{Z_0 + jZ_L \tan(\beta l)} \right)$$

$$Z_{in}(z) = Z_0 \left(\frac{1 + \Gamma(z)}{1 - \Gamma(z)} \right)$$

$$Z_{in} = jZ_0 \tan(\beta l)$$

$$Z_{in} = -jZ_0 \cot(\beta l)$$

$$\Gamma_L = |\Gamma_L| e^{j\phi}$$

$$|V(z)| = |A| \left| 1 + |\Gamma_L| e^{+j(\phi+2\beta z)} \right|$$

$$\left| \frac{V(z)}{V^+} \right| = \left| 1 + |\Gamma_L| e^{+j(\phi+2\beta z)} \right| = |1 + \Gamma(z)|$$

$$\Gamma(z) = \Gamma_L e^{j2\beta z} = \Gamma_L e^{-j2\beta l}$$

$$\text{VSWR} \equiv \frac{V_{\max}}{V_{\min}}$$

$$\text{VSWR} \equiv \frac{1+\left|\Gamma_{\textit{L}}\right|}{1-\left|\Gamma_{\textit{L}}\right|}$$

$$\text{SWR}=\max\left(\frac{R_L}{Z_0},\frac{Z_0}{R_L}\right)$$

$$Z_{in}^N(z)=\frac{1+\Gamma_{\textit{L}}\,e^{+2j\beta z}}{1-\Gamma_{\textit{L}}\,e^{+2j\beta z}}$$

$$Z_{in}=\frac{Z_{0\mathrm{T}}^2}{Z_L}$$

$$Z_{0\mathrm{T}}=\sqrt{Z_0R_L}$$

$$k=\omega\sqrt{\mu\varepsilon}$$

$$\eta=\sqrt{\frac{\mu}{\varepsilon}}$$

$$\eta_0=\sqrt{\frac{\mu_0}{\varepsilon_0}}\doteq 376.730313\;[\Omega]$$

$$\underline{S}=\hat{\underline{z}}\frac{\left|E_0\right|^2}{2\eta}$$

$$v_p=\frac{\omega}{k}$$

$$\lambda=\frac{2\pi}{k}$$

$$\lambda = \frac{c_d}{f}$$

$$E_x = E_0\,e^{-jkz}$$

$$H_y = \frac{1}{\eta} E_0\,e^{-jkz}$$

$$\varepsilon_c = \varepsilon - j\left(\frac{\sigma}{\omega}\right)$$

$$k=k'-jk''$$

$$\lambda = \frac{2\pi}{k'}$$

$$d_p=1/k''$$

$$\tan \delta = \frac{\varepsilon_c''}{\varepsilon_c'} = \frac{\sigma}{\omega \varepsilon}$$

$$\delta = d_p = \sqrt{\frac{2}{\omega \mu \sigma}}$$

$$k'\approx k''\approx \frac{1}{\delta}$$

$$Z_s \equiv \frac{E_{x0}}{J_{sx}}$$

$$R_s = \frac{1}{\sigma \delta} = \sqrt{\frac{\omega \mu}{2 \sigma}}$$

$$X_s=R_s$$

$$R = X = R_s \left(\frac{l}{2\pi a} \right)$$

$$R = R_s \left(\frac{1}{2\pi a} + \frac{1}{2\pi b} \right)$$

$$\underline{E}(z) = (\underline{\hat{x}} E_x + \underline{\hat{y}} E_y) e^{-jkz}$$

$$(a) \quad 0 < \beta < \pi \quad \text{LHEP}$$

$$(b) \quad -\pi < \beta < 0 \quad \text{RHEP}$$

$$\gamma = \tan^{-1} \left(\frac{b}{a} \right)$$

$$0 \leq \gamma \leq 90^\circ$$

$$\text{AR} = |\cot \xi|$$

$$\xi > 0: \quad \text{LHEP}$$

$$\xi < 0: \quad \text{RHEP}$$

where

$$\sin 2\xi = \sin 2\gamma \sin \beta$$

$$-45^\circ \leq \xi \leq +45^\circ$$

$$n_1 \sin \theta_i = n_2 \sin \theta_t$$

$$n_1 \sin \theta_c = n_2 \sin 90^\circ$$

$$\theta_i = \theta_b = \tan^{-1} \sqrt{\frac{\epsilon_2}{\epsilon_1}}$$

$$\Gamma_{TM} = \frac{Z_2^{TM} - Z_1^{TM}}{Z_2^{TM} + Z_1^{TM}}$$

$$\Gamma_{TE} = \frac{Z_2^{TE} - Z_1^{TE}}{Z_2^{TE} + Z_1^{TE}}$$

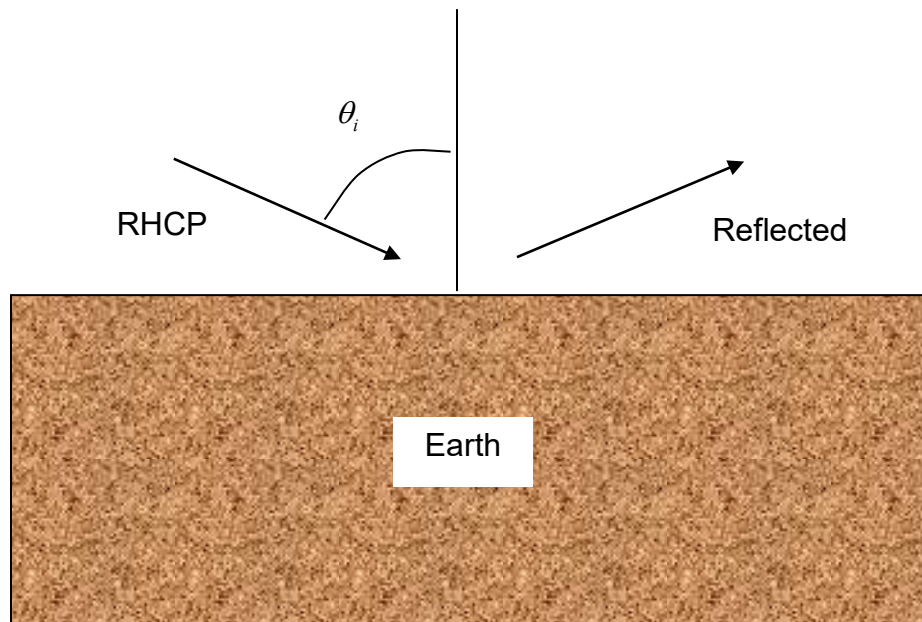
$$Z_1^{TE} = \left(\frac{\omega \mu_1}{k_{zi}} \right) \quad Z_2^{TE} = \left(\frac{\omega \mu_2}{k_{zt}} \right)$$

$$Z_1^{TM} = \left(\frac{k_{zi}}{\omega \varepsilon_1} \right) \quad Z_2^{TM} = \left(\frac{k_{zt}}{\omega \varepsilon_2} \right)$$

Problem 1 (30 pts)

A RHCP plane wave at 1.575 GHz from a GPS satellite is incident on the earth. The earth has a relative permittivity of $\epsilon_r = 6.0$ and is taken as lossless. The angle of incidence is $\theta_i = 60^\circ$. The power density in the incident plane wave from the GPS satellite (at an altitude of 20,200 km) is $10^{-13} \text{ [W/m}^2\text{]}$.

- (a) Calculate the power density in the reflected plane wave.
- (b) Calculate the percentage of the reflected power density that is in the TM_z polarization.
- (c) What would the angle of incidence have to be if we wanted no power to be reflected in the TM_z polarization?



Room for work

Answer

Part (a)

The reflection coefficients are

$$\Gamma_{TM} = -0.1339$$

$$\Gamma_{TE} = -0.6417 .$$

The power density is

$$P_d = P_{inc} \left(\frac{1}{2} \right) (|\Gamma_{TM}|^2 + |\Gamma_{TE}|^2),$$

where

$$P_{inc} = 10^{-13} \left[\text{W/m}^2 \right].$$

This gives

$$P_d = 2.149 \times 10^{-14} \left[\text{W/m}^2 \right].$$

Part (b)

We have

$$\%TM = \frac{\frac{1}{2} |\Gamma_{TM}|^2}{\frac{1}{2} (|\Gamma_{TM}|^2 + |\Gamma_{TE}|^2)} .$$

Hence, we have

$$\%TM = 4.17 .$$

Part (c)

The wave would be incident at the Brewster angle,

$$\theta_b = \tan^{-1} \sqrt{\frac{\varepsilon_2}{\varepsilon_1}}.$$

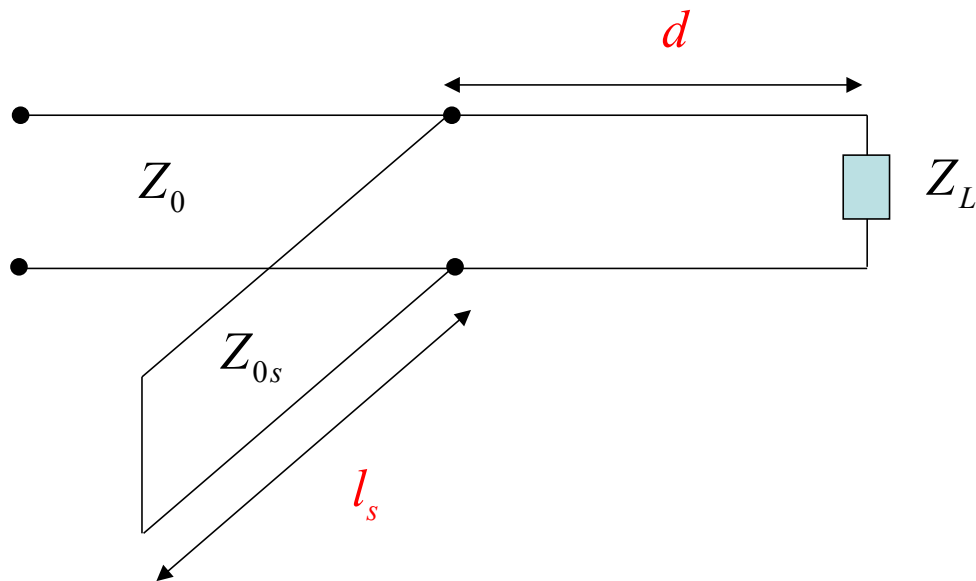
Hence, we have

$$\theta_b = 67.8^\circ.$$

Problem 2 (40 pts)

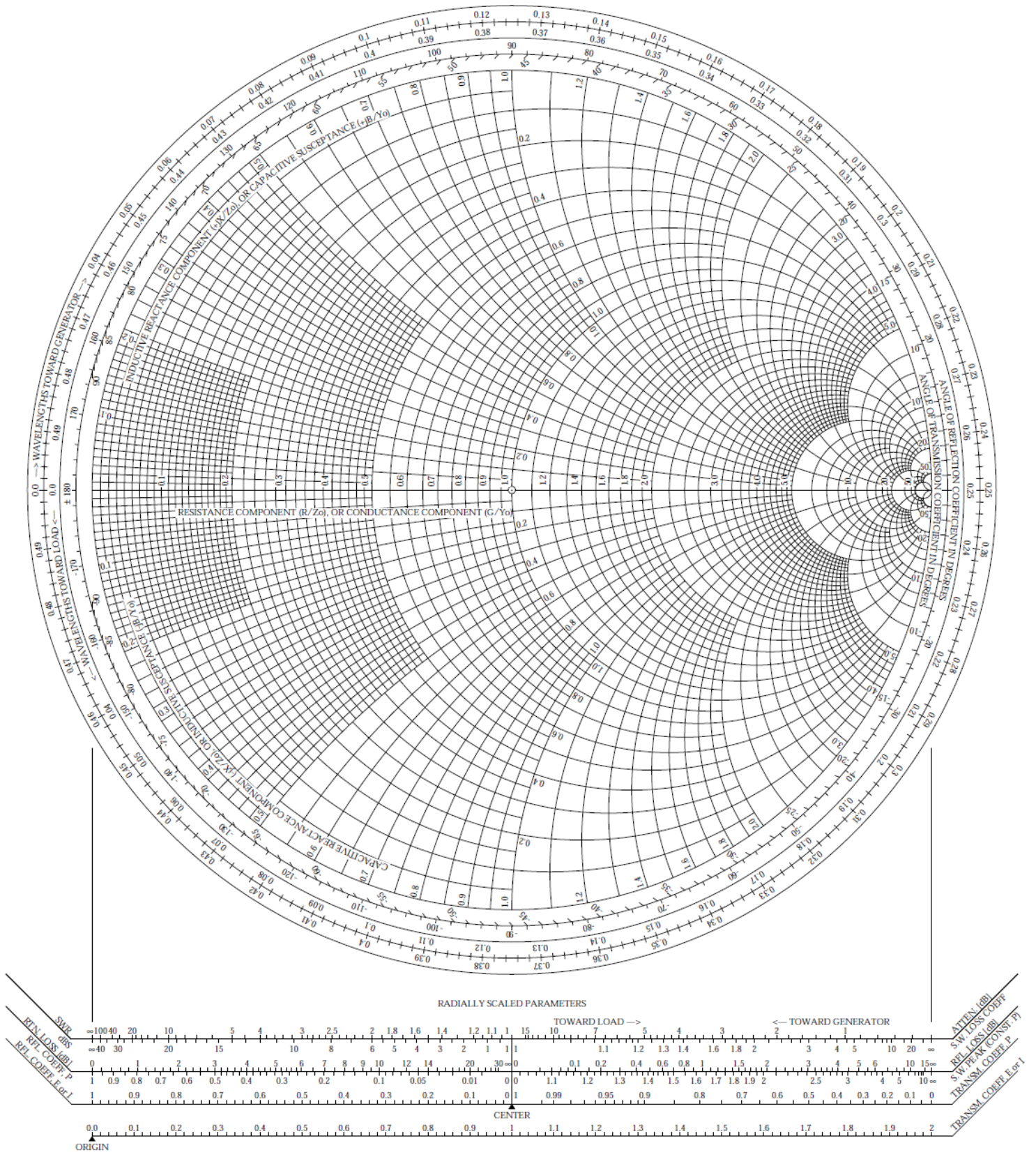
A twin-lead transmission line with $Z_0 = 300 \text{ } [\Omega]$ is connected to a load impedance that is given by $Z_L = 100 - j(100) \text{ } [\Omega]$. Assume that the frequency is 500 MHz and the effective permittivity of the line is 1.0 (the line is in air). In order to match the line to the load, a shorted-circuited stub with characteristic impedance $Z_{0s} = Z_0$ is inserted a distance d from the load.

Determine the distance d using the Smith chart that is on the next page (use the smallest d possible). Then find the length l_s of the stub line using the Smith chart that is attached.



The Complete Smith Chart

Black Magic Design



Room for work

Answer

The normalized load impedance is

$$Z_L^N = \frac{1}{3} - j\left(\frac{1}{3}\right).$$

The normalized admittance is

$$Y_L^N = \frac{1}{Z_L^N} = 1.5 + j(1.5).$$

We stay on a circle of constant radius, and rotate clockwise until we reach the circle $G_{in}^N = 1$. This corresponds to a distance d that is

$$d = 0.132\lambda_d.$$

At this point, the normalized input admittance is

$$Y_{in}^N = 1 - j(1.30).$$

Hence, for the stub we have

$$Y_s^N = +j(1.30).$$

We find the stub length, we start at the short-circuit position (the right most point on the real axis of the Smith chart when used as an admittance chart) and rotate clockwise until we reach this normalized admittance. This corresponds to a distance l_s that is

$$l_s = 0.396\lambda_d.$$

At the frequency of 500 [MHz] we have $\lambda_d = \lambda_0 = \frac{c}{f}$

so that

$$\lambda_d = 59.958 [\text{cm}].$$

Problem 3 (30 pts)

Consider the following plane wave that is traveling in air:

$$\underline{E} = \left[(1 - j3) \underline{\hat{y}} + (2 + j) \underline{\hat{z}} \right] e^{-jk_0 x}.$$

- (a) Find the polarization (linear, circular, or elliptical) and handedness (left-handed or right-handed) for the wave.
- (b) Find the axial ratio of this wave.
- (c) Find the magnetic field vector for this plane wave.

Room for work

Answer

Part (a)

We can rotate the coordinates so that we have

$$\underline{E} = \left[(1 - j3)\underline{\hat{x}} + (2 + j)\underline{\hat{y}} \right] e^{-jk_0 z}.$$

At $z = 0$ we have

$$E_x = 1 - j3$$

$$E_y = 2 + j.$$

Plotting these two points in the complex plane, we see that E_y leads E_x , but not by 90° . Also, the magnitudes are not equal. Thus we have

Polarization = LHEP.

Part (b)

We can factor out the $(1 - j3)$ term to write

$$\underline{E} = (1 - j3) \left[(1)\underline{\hat{x}} + \left(\frac{2 + j}{1 - j3} \right) \underline{\hat{y}} \right] e^{-jk_0 z}.$$

Hence, we can choose

$$a = 1$$

$$b = \frac{2 + j}{1 - j3} = 0.7071 e^{j(1.713)} = 0.7071 \angle 98.13^\circ,$$

so that

$$\beta = 1.713 \text{ [radians]} = 98.13^\circ.$$

We also have

$$\gamma = 35.264^\circ$$

$$\xi = 34.48^\circ.$$

This gives us

$$\text{AR} = 1.456.$$

Part (c)

Each electric field component of the plane wave has a corresponding magnetic field component, whose amplitude is related by η_0 .

Hence, using

$$\underline{E} = \left[(1 - j3) \underline{\hat{y}} + (2 + j) \underline{\hat{z}} \right] e^{-jk_0 x},$$

we have

$$\underline{H} = \frac{1}{\eta_0} \left[(1 - j3) \underline{\hat{z}} + (2 + j) (-\underline{\hat{y}}) \right] e^{-jk_0 x}.$$

ECE 3317

Applied Electromagnetic Waves

Exam 2
Dec. 3, 2020

Solution

General Information:

The exam is open-book and open-notes. You are not allowed to use any device that has communication functionality (laptop, cell phone, ipad, etc.), except for the computer that you use during for the exam, and this must not be used to communicate in any way with anyone other than the instructor during the exam.

Remember, you are bound by the UH Academic Honesty Policy during the exam!

Instructions:

- Please make sure that you have your camera active at all times during the exam.
- Be prepared to share your screen with the instructor at any time if requested to do so.
- The exam will be in three parts (three problems). For each one, you will have 25 minutes to do the problem, and 5 minutes to scan your solution, convert it to a pdf file, and upload it on Blackboard. Each exam problem will be placed on Blackboard on a page called "Exam 2" at the time you are to begin on each part (2:30 pm, 3:00 pm, 3:30 pm). The upload links for the different parts of the exam will also be on this page.
- If you need to ask any questions, please chat in private with the instructor (not to everyone!)
- If you have any problems with the upload of one of the exam parts, please notify the instructor by private chat immediately and email your pdf file to the instructor (djackson@uh.edu).
- When you create your pdf file for each exam part (problem), please name your file using the following convention: **Exam 2 Part 1 Smith Mary.pdf**. Please name your file exactly as shown. Put spaces between each word of the file name as shown. Do not put hyphens or underscores between the different words. Capitalize only the first letter of each word in the file name as shown.

- Show all of your work. No credit will be given if the work required to obtain the solutions is not shown.
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Problem 1 (30 pts)

A coaxial cable has an inner radius of $a = 0.5$ [mm] and an outer radius of $b = 2$ [mm]. The coax is filled with (nonmagnetic) Teflon having $\epsilon_r = 2.1$ and $\tan \delta = 0.001$. The conductors are made of copper, having a conductivity of $\sigma = 3.0 \times 10^7$ [S/m]. The copper conductors are nonmagnetic ($\mu = \mu_0$).

- a) Find the attenuation constant α in [nepers/m] at a frequency of 100 [MHz].
- b) Assume that we are now operating at a frequency where $\alpha = 0.05$ [nepers/m]. How far along the cable do we have to go so that the signal amplitude is 15 dB smaller than at the beginning?

Part (a)

$$R = 1.443 \text{ } [\Omega/\text{m}]$$

$$L = 2.773 \times 10^{-7} \text{ } [\text{H}/\text{m}]$$

$$G = 5.295 \times 10^{-5} \text{ } [\text{S}/\text{m}]$$

$$C = 8.427 \times 10^{-11} \text{ } [\text{F}/\text{m}]$$

This gives

$$\gamma = 0.014101 + j(3.037188) \text{ } [1/\text{m}]$$

Hence we have

$$\alpha = 0.014101 \text{ } [\text{np}/\text{m}]$$

Part (b)

$$0.05(8.686)z = 15$$

$$z = 34.54 \text{ } [\text{m}]$$

Problem 2 (35 pts)

A transmission line has a characteristic impedance of $50 \text{ } [\Omega]$. On the line the voltage maximum is measured as 4.0 [V] and the voltage minimum is measured as 2.0 [V] . The voltage minimum is measured to be at a distance of 0.15λ from the load.

Determine the unknown load impedance by using formulas (do not use the Smith chart).

Solution

$$Z_L = Z_0 \left(\frac{1 + \Gamma_L}{1 - \Gamma_L} \right)$$

where

$$\Gamma_L = |\Gamma_L| e^{j\phi}$$

with

$$|\Gamma_L| = \frac{\text{SWR} - 1}{\text{SWR} + 1} = \frac{2 - 1}{2 + 1} = \frac{1}{3}$$

and

$$\phi + 2\beta z_{\min} = -(2n + 1)\pi$$

or

$$\phi + 2 \left(\frac{2\pi}{\lambda} \right) z_{\min} = -(2n + 1)\pi$$

or

$$\phi = -4\pi \left(\frac{z_{\min}}{\lambda} \right) - (2n + 1)\pi$$

or

$$\phi = -1.257 \text{ [radians]} \text{ (using } n = 0 \text{ , which is an arbitrary choice).}$$

This gives us

$$\Gamma_L = \frac{1}{3}e^{-j1.257}$$

and

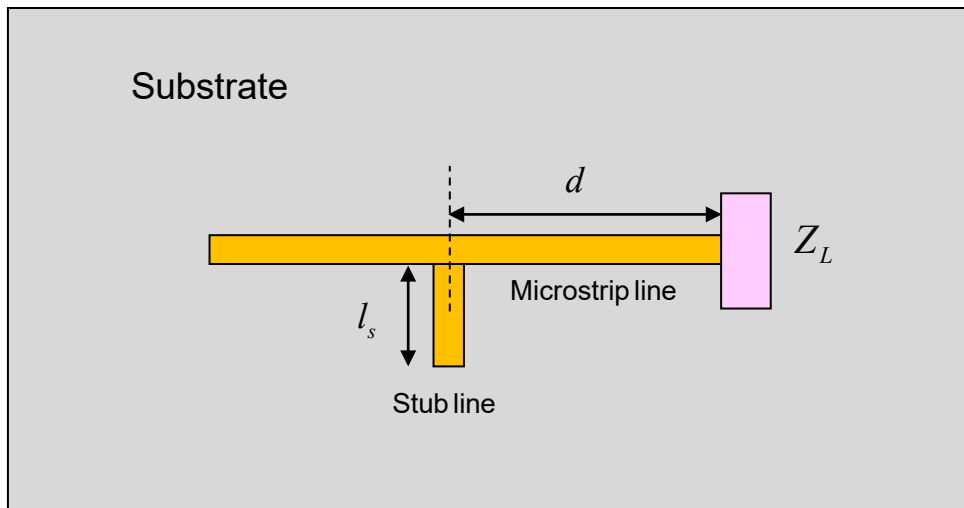
$$Z_L = 49.092 - j35.021 \quad [\Omega].$$

Problem 3 (35 pts)

A microstrip line has a characteristic impedance of $50\ [\Omega]$. The microstrip line meets a load impedance $Z_L = 100\ [\Omega]$. A stub line also has a characteristic impedance of $50\ [\Omega]$. This stub line is open-circuited at the end. It has a length of l_s and is placed at a distance of d from the load. The stub line is placed so that there is a perfect match on the main line to the left of the stub line. A top view is shown below.

Use the Smith chart to find d and l_s (in terms of the guided wavelength λ_g on the lines). Use the smallest value of d possible.

Clearly explain how you are using the Smith chart, and make sure that you attach your Smith chart showing your work. You may use the Smith chart on the next page, or one of your own.



Solution

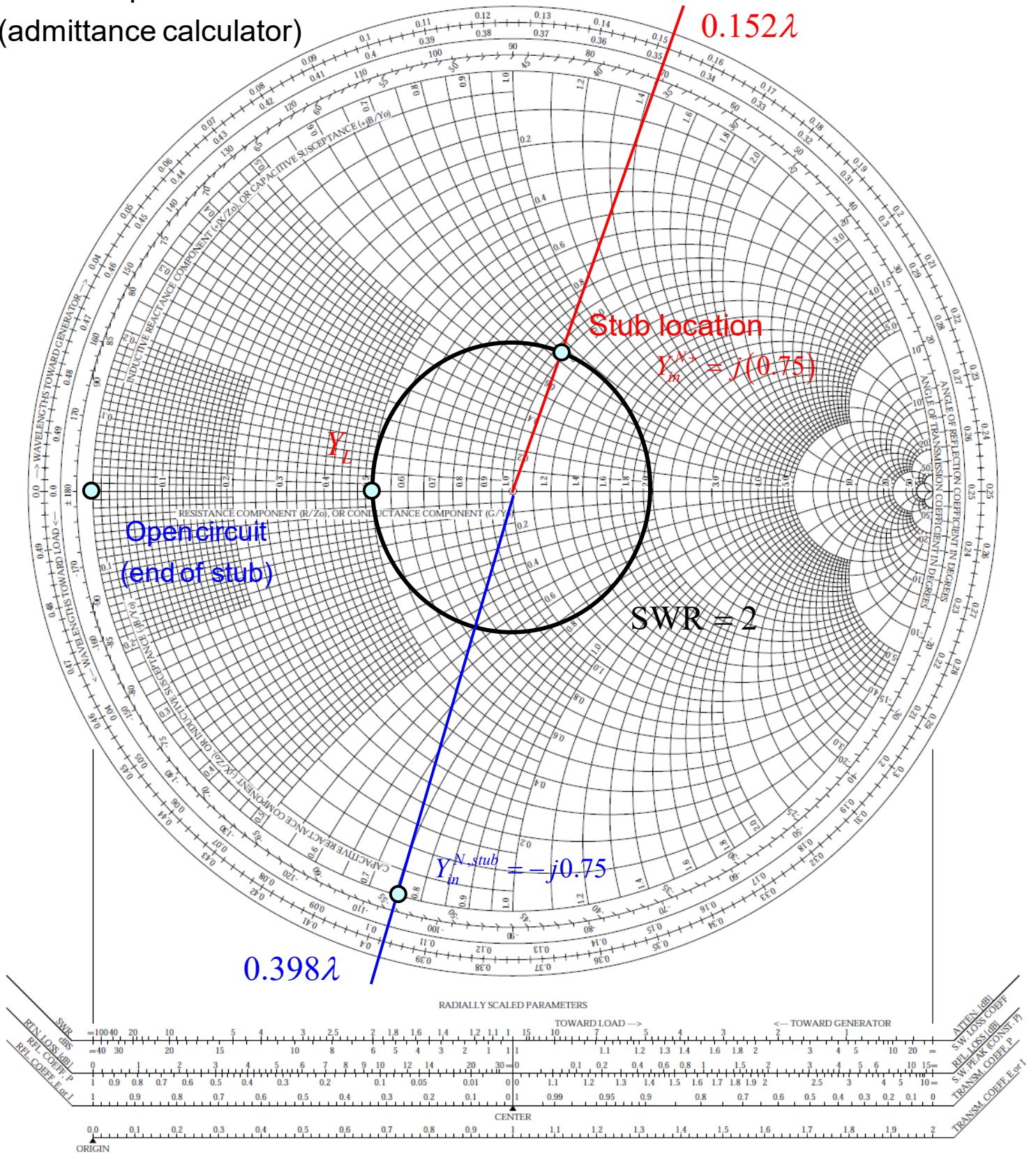
$$d = 0.152\lambda$$

$$l_s = 0.398\lambda$$

The Complete Smith Chart

Black Magic Design

Γ' plane
(admittance calculator)



Problem 3 (35 pts)

ECE 3317
Applied Electromagnetic Waves

Exam 2
Nov. 30, 2021

Name: _____ **SOLUTION** _____

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Signature: _____

Problem 1 (30 pts)

A coaxial cable has an inner radius of $a = 1.0$ [mm] and an outer radius of $b = 3.5$ [mm]. The coax is filled with (nonmagnetic) Teflon having $\epsilon_r = 2.25$ and $\tan \delta = 0.001$. The conductors are made of copper, having a conductivity of $\sigma = 3.0 \times 10^7$ [S/m]. The copper conductors are nonmagnetic ($\mu = \mu_0$).

- a) Find (R, L, G, C) for the coaxial cable at a frequency of 500 [MHz].
- b) Find the attenuation constant α in [nepers/m] at a frequency of 500 [MHz].
- c) Assume that at 1.0 GHz the attenuation on this line is now 0.5 [nepers/m]. What is the maximum length of cable that we can have if we want to keep the attenuation along the cable to be less than 10 dB?

Solution

Part (a)

$$R = 1.66 \text{ } [\Omega/\text{m}]$$

$$L = 2.506 \times 10^{-7} \text{ } [\text{H}/\text{m}]$$

$$G = 3.139 \times 10^{-4} \text{ } [\text{S}/\text{m}]$$

$$C = 9.992 \times 10^{-11} \text{ } [\text{F}/\text{m}]$$

Part (b)

$$\gamma = \sqrt{(R + j\omega L)(G + j\omega C)}$$

$$\gamma = 0.024433 + j(15.71884) \text{ } [1/\text{m}]$$

$$\alpha = 0.024433 \text{ } [\text{np}/\text{m}]$$

Part (c)

$$\alpha(8.686)d = 10 \text{ } [\text{dB}] \quad \rightarrow \quad 0.5(8.686)d = 10 \text{ } [\text{dB}]$$

$$d = 2.303 \text{ } [\text{m}]$$

Problem 2 (35 pts)

A coaxial cable transmission line has a characteristic impedance of $50\ [\Omega]$. The cable may be assumed to be lossless, and filled with a dielectric material having an unknown relative permittivity. The line is operating at 1.0 GHz.

On this line the magnitude of the voltage is maximum at a distance of 3.0 [cm] from the load, and the magnitude of the maximum voltage is measured as 3.0 [V]. The closest voltage minimum to this voltage maximum point is at a distance of 5.17 [cm] from the voltage maximum, and at this point the magnitude of the voltage is measured to be 1.0 [V].

- a) What is the relative permittivity of the dielectric that is filling the coax?
- b) Determine the unknown load impedance by using the Smith chart.

For part (b), clearly explain how you are using the Smith chart, and make sure that you attach your Smith chart showing your work. You may use the Smith chart below, or one of your own (it is preferred that you use the one below to reduce the risk of losing your Smith chart).

Solution

Part (a)

$$\frac{\lambda_g}{4} = \frac{\lambda_d}{4} = \frac{1}{4} \frac{\lambda_0}{\sqrt{\epsilon_r}} = 5.17 \times 10^{-2} \text{ [m]} , \quad \lambda_0 = \frac{c}{f} = 0.2998 \text{ [m]}$$

Hence,

$$\epsilon_r = 2.10$$

Part (b)

$$\text{SWR} = 3.0$$

$$\lambda_g = 20.68 \text{ [cm]} \rightarrow d_{\max} = 0.145 \lambda_g$$

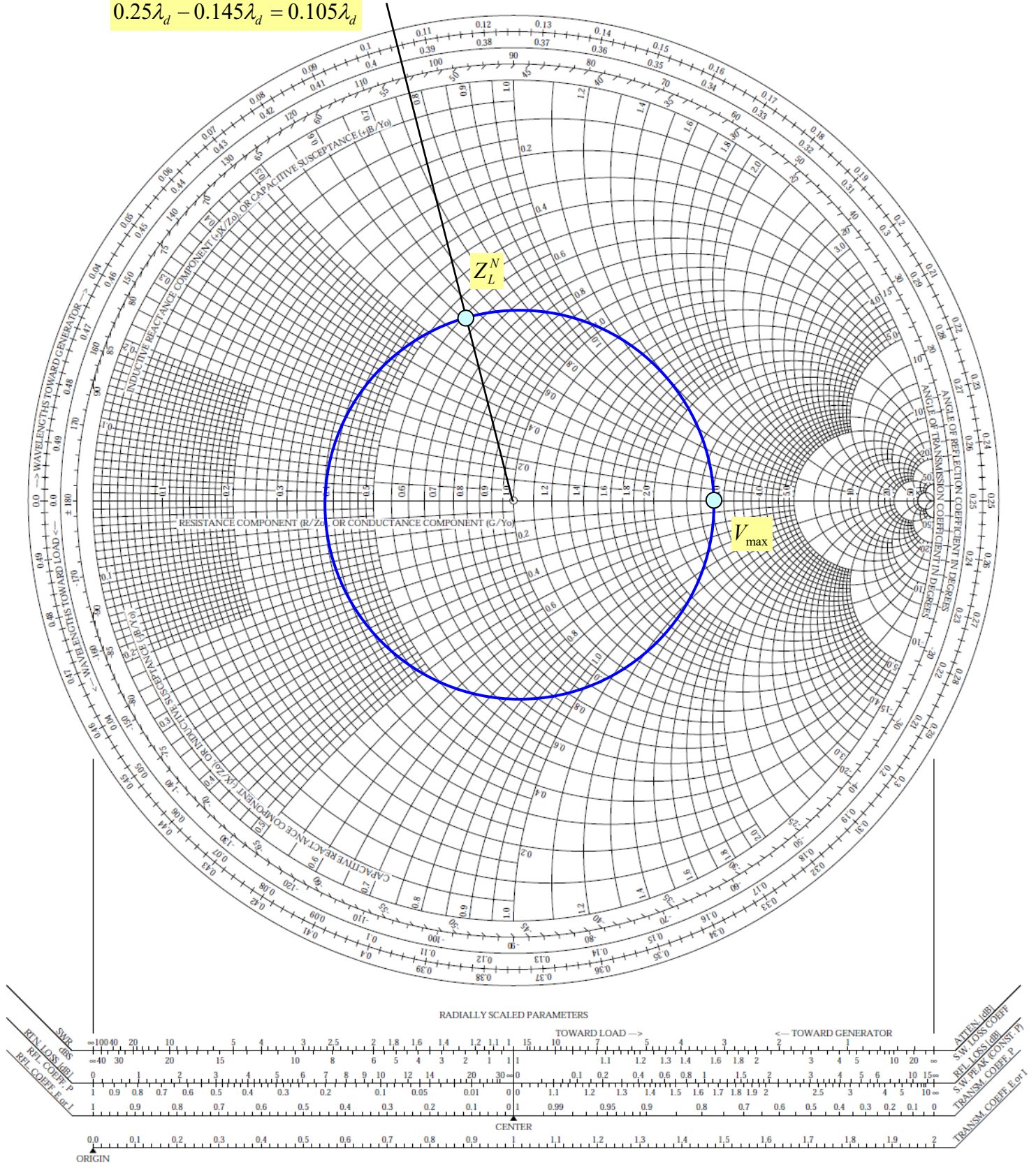
$$Z_L^N = 0.57 + j(0.62)$$

$$Z_L = 28.5 + j(31) \text{ } [\Omega]$$

The Complete Smith Chart

Black Magic Design

$$0.25\lambda_d - 0.145\lambda_d = 0.105\lambda_d$$

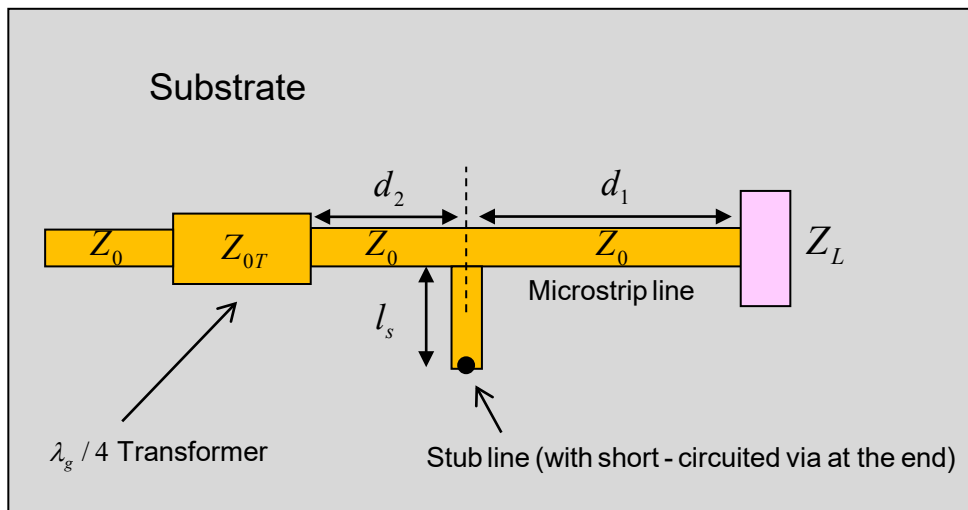


Problem 3 (35 pts)

A microstrip line of length d_1 has a characteristic impedance of $Z_0 = 50 \text{ } [\Omega]$. The microstrip line meets a load impedance $Z_L = 100 + j100 \text{ } [\Omega]$. The length d_1 is one-half of a guided wavelength. At the left end of this line a short-circuited stub line (also with a characteristic impedance of $50 \text{ } [\Omega]$) is placed in order to make the input impedance at this point purely real. A second line (also with a characteristic impedance of $50 \text{ } [\Omega]$) is then placed to the left of the stub, having a length d_2 that is one-fourth of a guided wavelength. To the left of this second line is a quarter-wave transformer, which transforms the impedance to $50 \text{ } [\Omega]$.

- Use the Smith chart to find the length l_s of the stub line (in terms of the guided wavelength λ_g on the lines). Use the smallest value of l_s possible.
- Find the characteristic impedance Z_{0T} of the transformer line.

Clearly explain how you are using the Smith chart, and make sure that you attach your Smith chart showing your work. You may use the Smith chart below, or one of your own (it is preferred that you use the one below to reduce the risk of losing your Smith chart).



Solution

Because $d_1 = \lambda_g / 2$, the impedance seen looking to the right of the stub line is

$$Z_L = 100 + j100 \text{ } [\Omega] .$$

The normalized impedance at this point is $Z_L^N = 2 + j2$.

The normalized admittance is $Y_L^N = 0.25 - j0.25$.

The stub will have

$$Y_{in}^N = +j0.25 .$$

This corresponds to a stub length of

$$l_2 = 0.289 \lambda_g .$$

To the left of the stub we then have

$$Y_{in}^N = 0.25$$

so

$$R_{in}^N = 4.0 .$$

After the quarter-wave section, we then have

$$R_{in}^N = 0.25 .$$

The load seen by the transformer is then

$$R_{in} = 12.5 \text{ } [\Omega] .$$

We then have

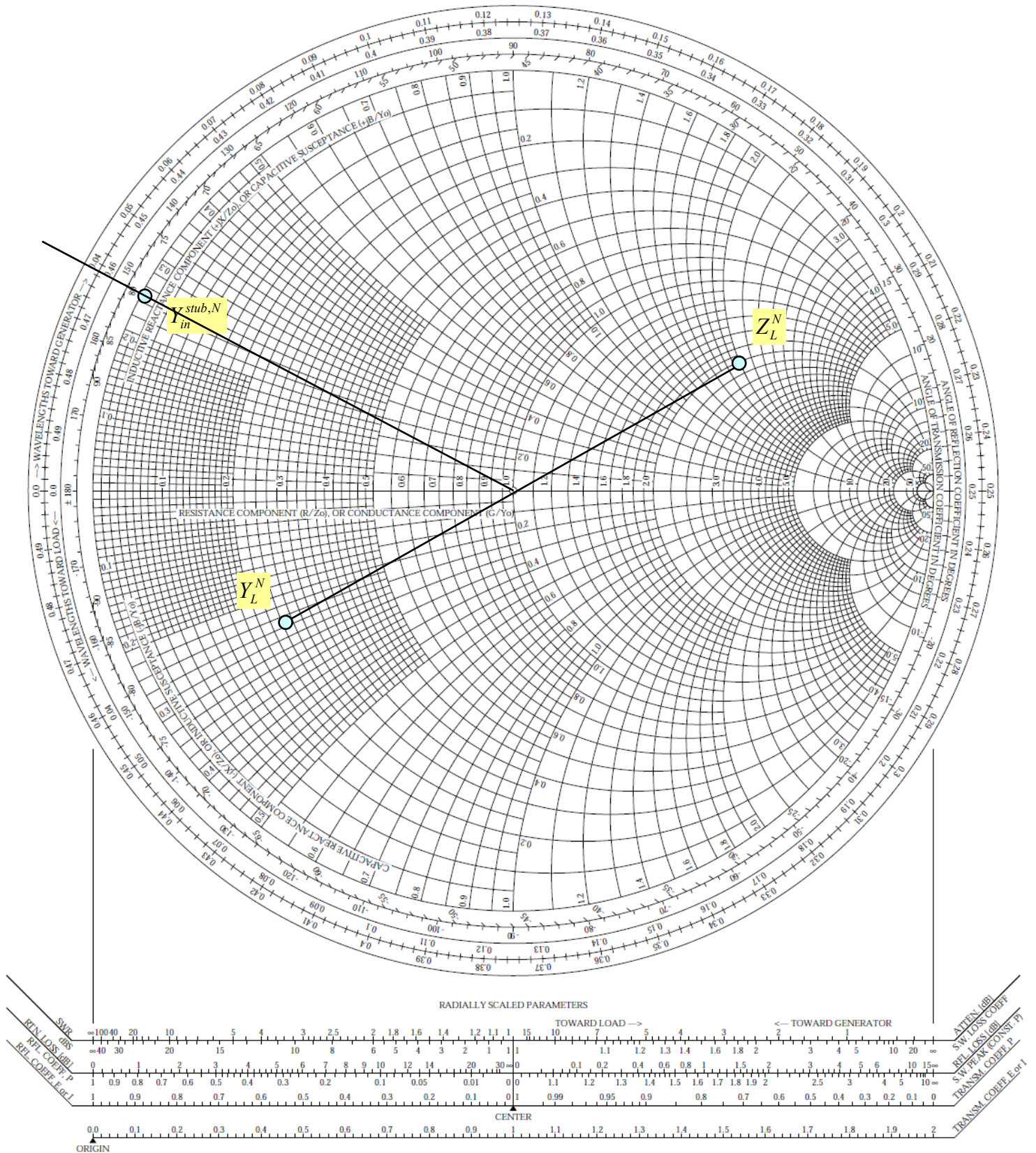
$$Z_{0T} = \sqrt{(50)(12.5)}$$

so that

$$Z_{0T} = 25.0 \text{ } [\Omega]$$

The Complete Smith Chart

Black Magic Design



ECE 3317
Applied Electromagnetic Waves

Exam 2
Dec. 1, 2022

Name: _____ **SOLUTION** _____

General Information:

The exam is open-book and open-notes. You are not allowed to use any device that has communication functionality (laptop, cell phone, ipad, etc.).

Remember, you are bound by the UH Academic Honesty Policy during the exam!

Instructions:

- Show all of your work. No credit will be given if the work required to obtain the solutions is not shown.
- Write neatly. You will not be given credit for work that is not easily legible.
- Leave answers in terms of the parameters given in the problem.
- Show units in all of your final answers.
- Circle your final answers.
- Double-check your answers. For simpler problems, partial credit may not be given.
- If you have any questions, ask the instructor. You will not be given credit for work that is based on a wrong assumption.
- Make sure you sign the academic honesty statement below.

Academic Honesty Statement

By taking this exam, you agree to abide by the UH Academic Honesty Policy during this exam. You understand and agree that the punishment for violating this policy will be most severe, including getting an F in the class and getting expelled from the University.

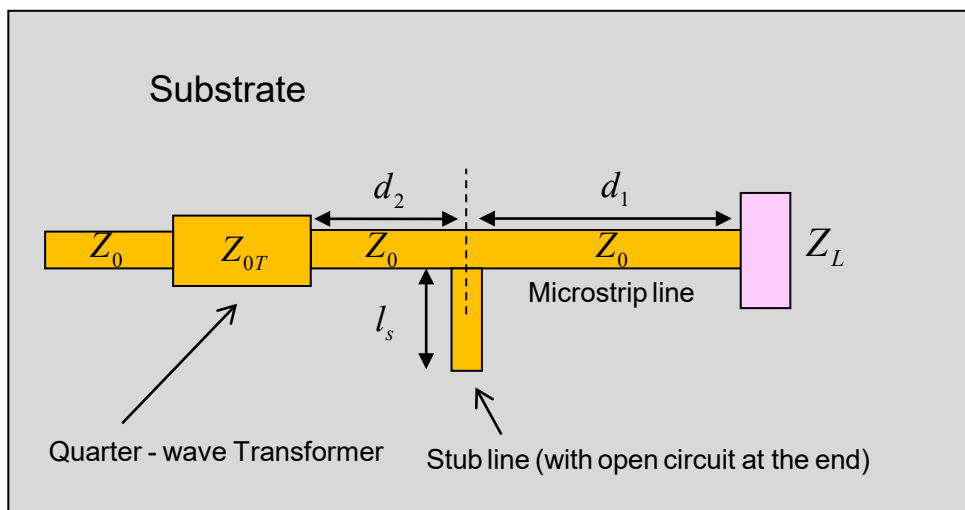
Signature: _____

Problem 1 (35 pts)

A microstrip line of length d_1 has a characteristic impedance of $Z_0 = 50 \text{ } [\Omega]$. The length d_1 is 0.1 of a guided wavelength on this line. The microstrip line meets a load impedance $Z_L = 75 - j50 \text{ } [\Omega]$. At the left end of this line an open-circuited stub line (also with a characteristic impedance of $50 \text{ } [\Omega]$) is placed. To the left of this stub line is another section of $50 \text{ } [\Omega]$ line that has a length d_2 that is one guided wavelength on this line. To the left of this line is a quarter-wave transformer, which transforms the impedance to $50 \text{ } [\Omega]$, so that the incoming feed line to the left of the transformer sees a match. A top view is shown below.

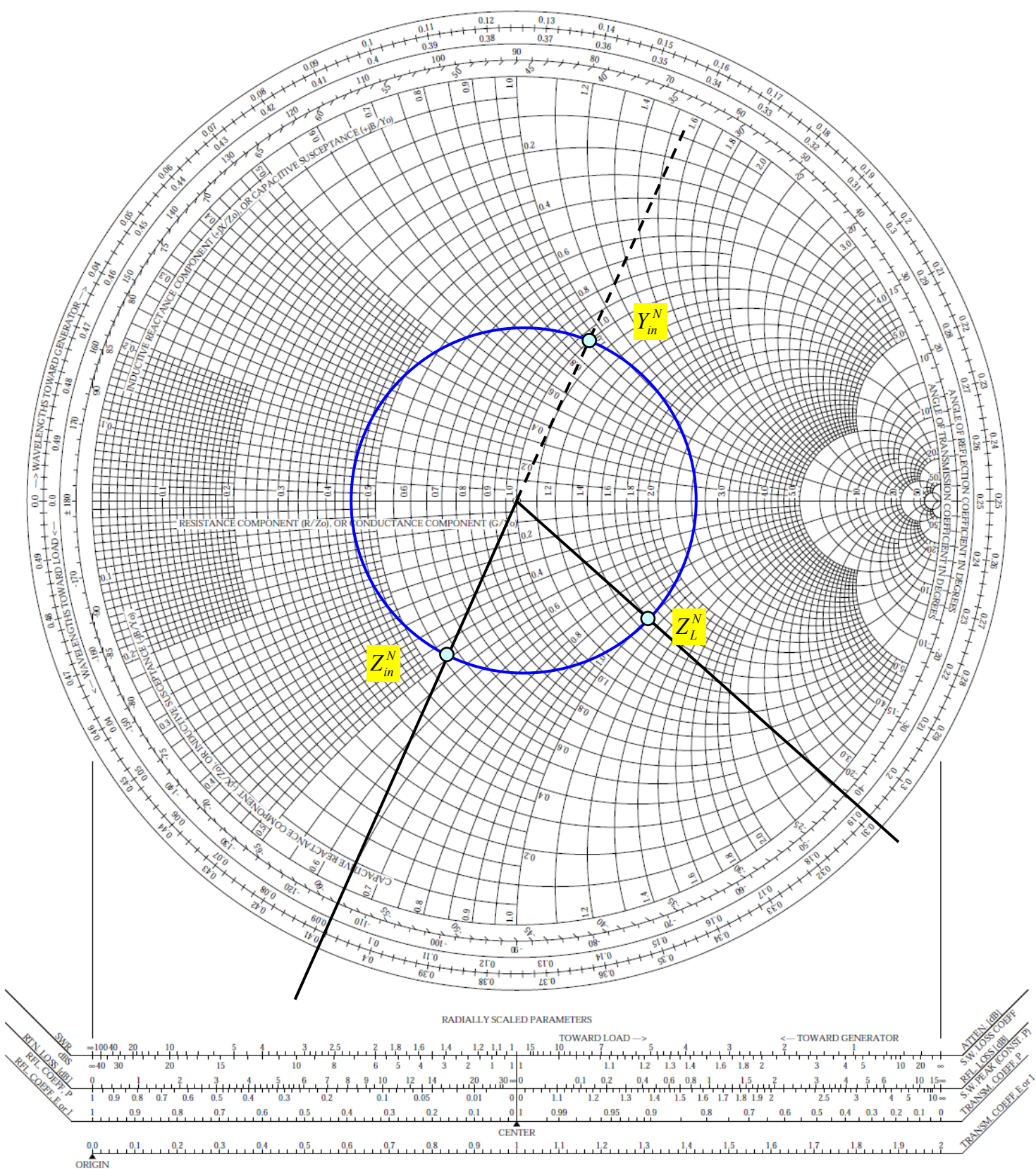
- Use the first Smith chart below to find the input impedance (in Ohms) seen looking to the right, at a point that is just to the right of the stub line.
- Use the same first Smith chart to find the SWR on the line that is connected to the load.
- Use the second Smith chart below to find the length l_s of the stub line (in terms of the guided wavelength λ_g on the $50 \text{ } [\Omega]$ stub line). Use the smallest value of l_s possible.
- Find the characteristic impedance Z_{0T} of the transformer line.

Clearly explain how you are using the Smith charts.



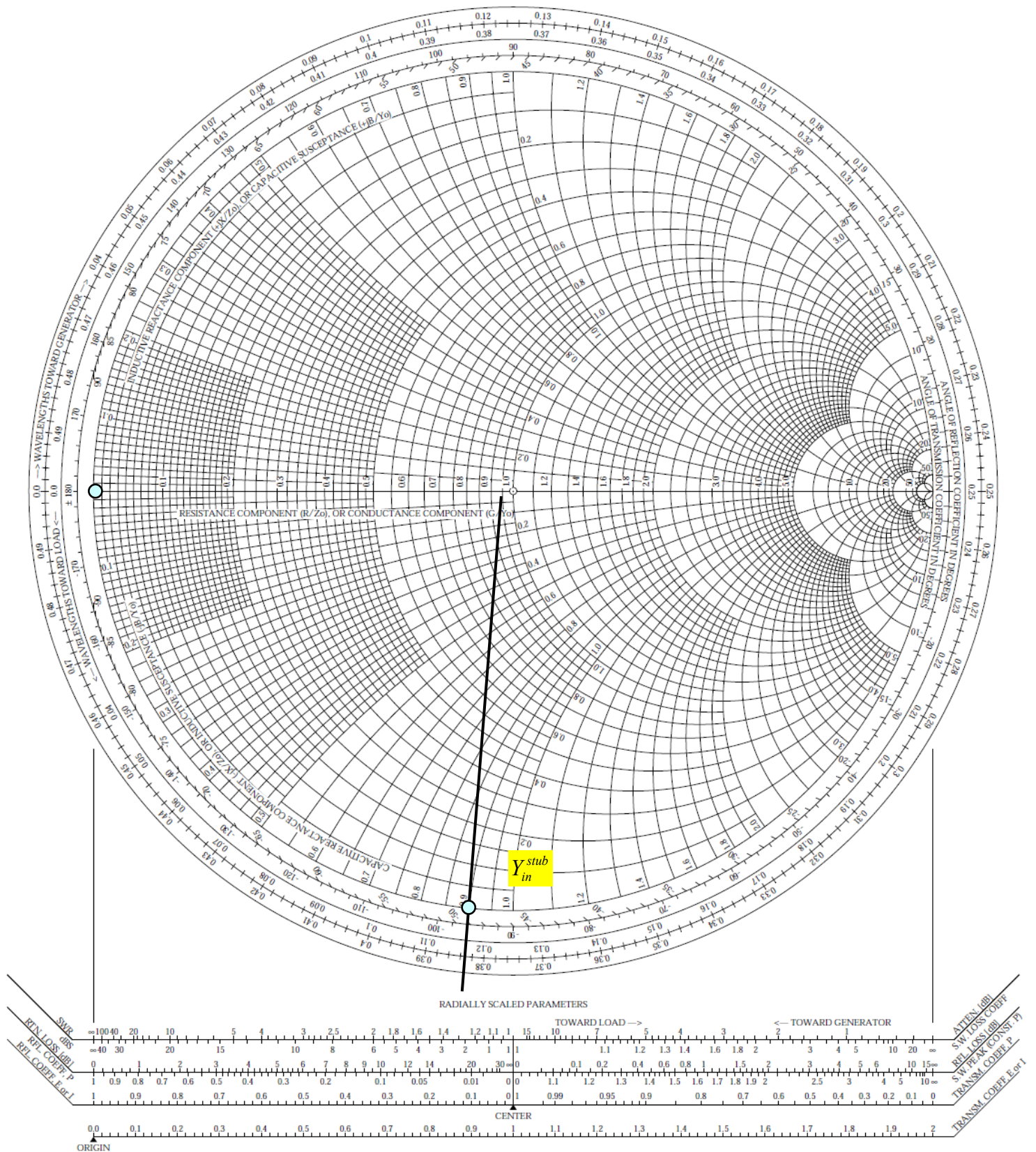
The Complete Smith Chart

Black Magic Design



The Complete Smith Chart

Black Magic Design



SOLUTION

Part (a)

From the Smith chart,

$$Z_{in}^N = 0.57 - j(0.48) .$$

Hence,

$$Z_{in}^N = 28.5 - j(24.0) .$$

Part (b)

From where the SWR circle intersects the positive real axis, we see that

$$\text{SWR} = 2.45 .$$

Part (c)

From the Smith chart (going halfway around), we have

$$Y_{in}^N = 1.00 + j(0.90) .$$

Therefore,

$$Y_{in}^{stub} = -j(0.90) .$$

From the second Smith chart we then have

$$l_s = 0.383\lambda_g .$$

Part (d)

We have, just to the left of the stub line,

$$Y_{in}^N = 1.00$$

so

$$Z_{in} = 50.0 \, [\Omega].$$

We then have

$$Z_{0T} = \sqrt{(50)(50.0)} .$$

We then have

$$Z_{0T} = 50.0 \, [\Omega].$$

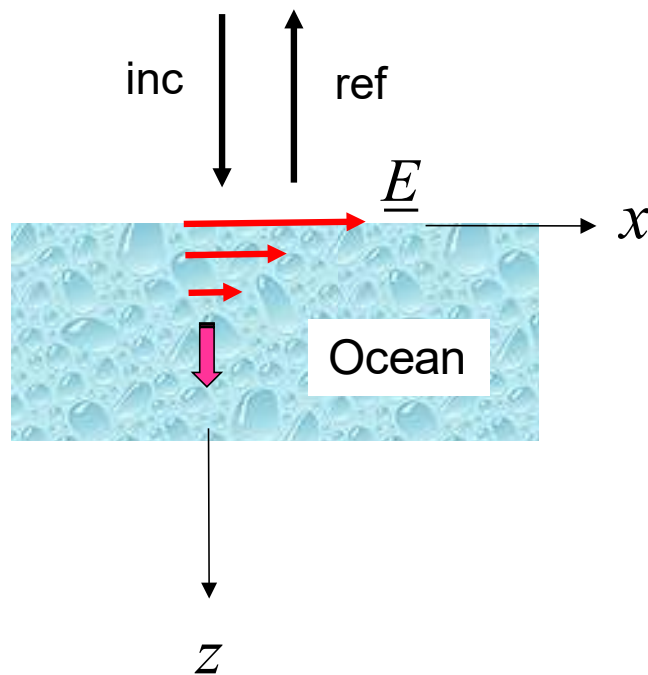
Note that in this case, the transformer is not really needed!

Problem 2 (35 pts)

A plane wave in air is incident vertically on the ocean at 18 GHz. The plane wave is polarized with the electric field in the x direction. The relative permittivity of the ocean water at this frequency is $\epsilon_r = 38 - j38$. The ocean water also has a conductivity of $\sigma = 4$ [S/m]. The ocean water is nonmagnetic.

- Find the complex relative effective permittivity ϵ_{re} of the ocean water.
- Find the loss tangent of the ocean water.
- Find the attenuation in [dB/m] in the ocean water.
- Find the percentage of power that gets reflected from the surface of the ocean.

Note: In part (d) it doesn't matter if you take the wave to be TM or TE, since they are the same for vertical incidence.



SOLUTION

Part (a)

We have

$$\varepsilon_c = \varepsilon - j\left(\frac{\sigma}{\omega}\right) = \varepsilon_0 \left[\varepsilon_r - j\left(\frac{\sigma}{\omega\varepsilon_0}\right) \right]$$

so

$$\varepsilon_{rc} = \varepsilon_r - j\left(\frac{\sigma}{\omega\varepsilon_0}\right).$$

This gives us

$$\varepsilon_{rc} = 38 - j(41.994).$$

Part (b)

The loss tangent is

$$\tan \delta = \frac{\varepsilon''}{\varepsilon'} = \frac{-\operatorname{Im}(\varepsilon_c)}{\operatorname{Re}(\varepsilon_c)} = \frac{-\operatorname{Im}(\varepsilon_{rc})}{\operatorname{Re}(\varepsilon_{rc})}.$$

Hence, we have

$$\tan \delta = 1.105.$$

Part (c)

The wavenumber of the seawater is

$$k = \omega\sqrt{\mu_0\varepsilon_c} = \omega\sqrt{\mu_0\varepsilon_0\varepsilon_{rc}} = k_0\sqrt{\varepsilon_{rc}} = \frac{2\pi}{\lambda_0}\sqrt{\varepsilon_{rc}}.$$

This gives us

$$k = 2595.03 - j(1151.55)$$

The attenuation is thus

$$\alpha = 1151.55 \text{ [np/m]}.$$

Multiplying by 8.686, we then have

$$\text{Attenuation} = 10002.4 \text{ [dB/m]}$$

Part (d)

Because of the vertical incidence ($\theta_i = 0$), we have

$$Z_1^{\text{TM}} = Z_1^{\text{TE}} = \eta_1 = \eta_0 = 377.7303$$

$$Z_2^{\text{TM}} = Z_2^{\text{TE}} = \eta_2 = \frac{\eta_0}{\sqrt{\epsilon_{rc}}}.$$

Hence,

$$\Gamma = \frac{\frac{1}{\sqrt{\epsilon_{rc}}} - 1}{\frac{1}{\sqrt{\epsilon_{rc}}} + 1}.$$

This gives us

$$\Gamma = -0.779283 + j(0.085512).$$

The percent power reflected is

$$P_r^{\%} = |\Gamma|^2 100.$$

This gives us

$$P_r^{\%} = 61.46.$$

That is, 61.46% of the incident power is reflected.

Problem 3 (30 pts)

A plane wave in air has the following electric field:

$$\underline{E} = \left[(1 + j) \underline{\hat{x}} + (1 - j2) \underline{\hat{y}} \right] e^{-jk_0 z}.$$

- (a) Classify the polarization of this wave (linear, LHCP, RHCP, LHEP, RHEP).
- (b) Find the axial ratio of this wave.
- (c) Find the magnetic field for this wave.

SOLUTION

Part (a)

At $z = 0$ we have

$$E_x = (1 + j), \quad E_y = (1 - j2).$$

Normalizing (dividing by $1 + j$), we can write

$$E_x = 1, \quad E_y = -0.5 - j(1.5)$$

or

$$E_x = 1, \quad E_y = 1.58114 \angle -108.435^\circ.$$

Hence E_y lags E_x . In time, the wave therefore rotates from the x axis to the y axis. The wave propagates in the $+z$ direction. Hence, the wave is left-handed. Since the magnitude of E_y and E_x are not equal, and the phase difference is not $\pm 90^\circ$, the wave is not circularly-polarized.

Hence, we have

RHEP.

Part (b)

We have

$$\gamma = 57.69^\circ$$

$$\beta = -108.435^\circ.$$

This gives us

$$\xi = -29.499^\circ.$$

We have

$$\text{AR} = |\cot(\xi)|.$$

From this we get

$$\text{AR} = 1.768.$$

Part (c)

The magnetic field is found the components of the electric field, noting that each component of the magnetic field comes from the corresponding perpendicular component of the electric field after dividing by given by η_0 . In particular,

$$\frac{E_x}{H_y} = \eta_0, \quad \frac{E_y}{H_x} = -\eta_0.$$

We then have

$$\underline{H} = \frac{1}{\eta_0} \left[(1+j) \underline{\hat{y}} - (1-j2) \underline{\hat{x}} \right] e^{-jk_0 z}.$$

ROOM FOR WORK

DO NOT BEGIN THIS EXAM UNTIL TOLD TO START

Name: _____

PeopleSoft ID: _____

ECE 3317
Applied Electromagnetic Waves

Exam II
April 25, 2013

1. This exam is open book and open notes. However, you are not allowed to use a computer or any electronic device other than a calculator. Any devices that may be used to communicate are not allowed.
2. Show all of your work. No credit will be given if the work required to obtain the solutions is not shown.
3. Perform all your work on the exam in the space allowed.
4. Write neatly. You will not be given credit for work that is not **easily legible**.
5. Leave answers in terms of the parameters given in the problem.
6. Show units in all of your final answers.
7. Circle your final answers.
8. Double-check your answers. For simpler problems, partial credit may not be given.
9. If you have any questions, ask the instructor. You will not be given credit for work that is based on a wrong assumption.
10. Make sure you sign the academic honesty statement on the next page.

Academic Honesty Statement

I agree to abide by the UH Academic Honesty Policy during this exam. I understand that the punishment for violating this policy will be most severe, including getting an F in the class and getting expelled from the University.

Signature

FORMULA SHEET

$$\nabla \times \underline{\mathcal{E}} = -\frac{\partial \underline{\mathcal{B}}}{\partial t}$$

$$\nabla \times \underline{\mathcal{H}} = \underline{\mathcal{J}} + \frac{\partial \underline{\mathcal{Q}}}{\partial t}$$

$$\nabla \cdot \underline{\mathcal{B}} = 0$$

$$\nabla \cdot \underline{\mathcal{Q}} = \rho_v$$

$$\nabla \times \underline{E} = -j\omega \underline{B}$$

$$\nabla \times \underline{H} = \underline{J} + j\omega \underline{D}$$

$$\nabla \cdot \underline{B} = 0$$

$$\nabla \cdot \underline{D} = \rho_v$$

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$$

$$\varepsilon_0 \doteq 8.8541878 \times 10^{-12} \text{ [F/m]}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ [H/m]}$$

$$\oint_s (\underline{\mathcal{E}} \times \underline{\mathcal{H}}) \cdot \hat{n} dS = - \int_V \sigma |\underline{\mathcal{E}}|^2 dV - \int_V \frac{\partial}{\partial t} \left(\frac{1}{2} \mu |\underline{\mathcal{H}}|^2 \right) dV - \int_V \frac{\partial}{\partial t} \left(\frac{1}{2} \varepsilon |\underline{\mathcal{E}}|^2 \right) dV$$

$$\underline{\mathcal{J}} = \underline{\mathcal{E}} \times \underline{\mathcal{H}}$$

$$\underline{S} \equiv \frac{1}{2} (\underline{E} \times \underline{H}^*)$$

$$C = \frac{2\pi\epsilon_0\epsilon_r}{\ln\left(\frac{b}{a}\right)} \quad [\text{F/m}]$$

$$L = \frac{\mu_0}{2\pi} \ln\left(\frac{b}{a}\right) \quad [\text{H/m}]$$

$$G = \frac{2\pi\sigma_d}{\ln\left(\frac{b}{a}\right)} \quad [\text{S/m}]$$

$$R = \frac{1}{\sigma_m\delta} \left(\frac{1}{2\pi a} + \frac{1}{2\pi b} \right) \quad [\Omega/\text{m}]$$

$$\frac{\partial v}{\partial z} = -Ri - L \frac{\partial i}{\partial t}$$

$$\frac{\partial I}{\partial z} i = -Gv - C \frac{\partial v}{\partial t}$$

$$v(z,t) = f(z - c_d t) + g(z + c_d t)$$

$$i(z,t) = \frac{1}{Z_0} [f(z - c_d t) - g(z + c_d t)]$$

$$v(z,t) = v_g(t - z / c_d)$$

$$\Gamma_g = \left(\frac{R_g - Z_0}{R_g + Z_0} \right) \qquad \Gamma_L = \left(\frac{R_L - Z_0}{R_L + Z_0} \right)$$

$$V^+ = \left(\frac{Z_0}{R_g + Z_0} \right) V_0$$

$$\Gamma_L(t)=1-2e^{-(t-T)/\tau}, \quad t \geq T \qquad \tau = Z_0 C_L$$

$$\Gamma_L(t)=-1+2e^{-(t-T)/\tau}, \quad t \geq T \qquad \tau = L_L / Z_0$$

$$V(z)=Ae^{-\gamma z}+Be^{+\gamma z}$$

$$\gamma=\sqrt{(R+j\omega L)(G+j\omega C)}$$

$$LC=\mu\varepsilon=\frac{1}{c_d^2}$$

$$\gamma=\alpha+j\beta$$

$$k_z=-j\gamma=\beta-j\alpha$$

$$v_p=\frac{\omega}{\beta}$$

$$\beta=\frac{2\pi}{\lambda}$$

$$\text{attenuation}=\left(\frac{20}{\ln 10}\right)\alpha=(8.6859)\alpha \quad [\text{dB/m}]$$

$$Z_0=\sqrt{\frac{R+j\omega L}{G+j\omega C}}$$

$$i(z) = \left(\frac{1}{Z_0} \right) \left[A e^{-\gamma z} - B e^{+\gamma z} \right]$$

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$V(z) = A \left(e^{-\gamma z} + \Gamma_L e^{+\gamma z} \right)$$

$$I(z) = \frac{1}{Z_0} A \left(e^{-\gamma z} - \Gamma_L e^{+\gamma z} \right)$$

$$Z_{in}(z) = \frac{V(z)}{I(z)}$$

$$Z_{in}(z) = Z_0 \left(\frac{1 + \Gamma_L e^{+2\gamma z}}{1 - \Gamma_L e^{+2\gamma z}} \right)$$

$$Z_{in} = Z_0 \left(\frac{Z_L + Z_0 \tanh(\gamma l)}{Z_0 + Z_L \tanh(\gamma l)} \right)$$

$$Z_{in} = Z_0 \left(\frac{Z_L + jZ_0 \tan(\beta l)}{Z_0 + jZ_L \tan(\beta l)} \right)$$

$$Z_{in}(z) = Z_0 \left(\frac{1 + \Gamma(z)}{1 - \Gamma(z)} \right)$$

$$Z_{in} = jZ_0 \tan(\beta l)$$

$$Z_{in} = -jZ_0 \cot(\beta l)$$

$$\Gamma_L = |\Gamma_L| e^{j\phi}$$

$$|V(z)| = |A| \left| 1 + |\Gamma_L| e^{+j(\phi+2\beta z)} \right|$$

$$\left| \frac{V(z)}{V^+} \right| = \left| 1 + |\Gamma_L| e^{+j(\phi+2\beta z)} \right| = \left| 1 + \Gamma(z) \right|$$

$$\Gamma(z) = \Gamma_L e^{j2\beta z} = \Gamma_L e^{-j2\beta l}$$

$$\text{VSWR} \equiv \frac{V_{\max}}{V_{\min}}$$

$$\text{VSWR} \equiv \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|}$$

$$\text{SWR} = \max\left(\frac{R_L}{Z_0}, \frac{Z_0}{R_L}\right)$$

$$Z_{in}^N(z) = \frac{1 + \Gamma_L e^{+2j\beta z}}{1 - \Gamma_L e^{+2j\beta z}}$$

$$Z_{in} = \frac{Z_{0T}^2}{Z_L}$$

$$Z_{0T} = \sqrt{Z_0 R_L}$$

$$k=\omega\sqrt{\mu\varepsilon}$$

$$\eta=\sqrt{\frac{\mu}{\varepsilon}}$$

$$\eta_0=\sqrt{\frac{\mu_0}{\varepsilon_0}}\doteq 376.730313\;[\Omega]$$

$$\underline{S}=\underline{\hat{z}}\frac{\left|E_0\right|^2}{2\eta}$$

$$v_p=\frac{\omega}{k}$$

$$\lambda=\frac{2\pi}{k}$$

$$\lambda=\frac{c_d}{f}$$

$$E_x=E_0\,e^{-jkz}$$

$$H_y=\frac{1}{\eta}E_0\,e^{-jkz}$$

$$\varepsilon_c=\varepsilon-j\left(\frac{\sigma}{\omega}\right)$$

$$k=k'-jk''$$

$$\lambda = \frac{2\pi}{k'}$$

$$d_p = 1 / k''$$

$$\tan \delta = \frac{\varepsilon_c''}{\varepsilon_c'} = \frac{\sigma}{\omega \varepsilon}$$

$$\delta = d_p = \sqrt{\frac{2}{\omega \mu \sigma}}$$

$$k' \approx k'' \approx \frac{1}{\delta}$$

$$Z_s \equiv \frac{E_{x0}}{J_{sx}}$$

$$R_s = \frac{1}{\sigma \delta} = \sqrt{\frac{\omega \mu}{2 \sigma}}$$

$$X_s = R_s$$

$$R=X=R_s\left(\frac{l}{2\pi a}\right)$$

$$R=R_s\left(\frac{1}{2\pi a}+\frac{1}{2\pi b}\right)$$

$$\underline{E}(z) = (\hat{x} \, E_x + \hat{y} \, E_y) e^{-jkz}$$

$$(a) \quad 0 < \beta < \pi \quad \text{LHEP}$$

$$(b) \quad -\pi < \beta < 0 \quad \text{RHEP}$$

$$n_1 \sin \theta_i = n_2 \sin \theta_t$$

$$\Gamma_{TM} = \frac{Z_2^{TM} - Z_1^{TM}}{Z_2^{TM} + Z_1^{TM}}$$

$$\Gamma_{TE} = \frac{Z_2^{TE} - Z_1^{TE}}{Z_2^{TE} + Z_1^{TE}}$$

$$Z_1^{TE} = \left(\frac{\omega \mu_1}{k_{zi}} \right) \quad Z_2^{TE} = \left(\frac{\omega \mu_2}{k_{zt}} \right)$$

$$Z_1^{TM} = \left(\frac{k_{zi}}{\omega \varepsilon_1} \right) \quad Z_2^{TM} = \left(\frac{k_{zt}}{\omega \varepsilon_2} \right)$$

$$\theta_i = \theta_b = \tan^{-1} \sqrt{\frac{\varepsilon_2}{\varepsilon_1}}$$

$$k_z = k \sqrt{1 - (f_c / f)^2}$$

$$f_c = \frac{c_d}{2\pi} \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}$$

$$\beta = k \sqrt{1 - (f_c / f)^2}, \quad f > f_c$$

$$\alpha = k_c \sqrt{1 - (f / f_c)^2}, \quad f < f_c$$

$$H_z(x, y, z) = A_{10} \cos\left(\frac{\pi x}{a}\right) e^{-jk_z z}$$

$$\lambda_g = \frac{\lambda}{\sqrt{1 - (f_c / f)^2}}$$

$$v_p = \frac{c_d}{\sqrt{1 - (f_c / f)^2}}$$

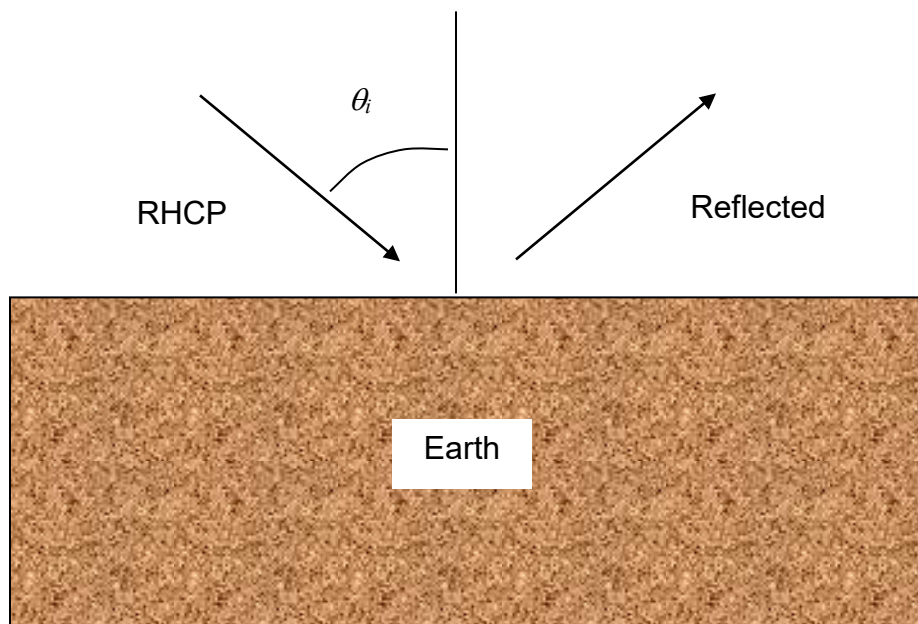
$$v_g = \frac{d\omega}{d\beta} = \frac{1}{\frac{d\beta}{d\omega}}$$

$$v_g = c_d \sqrt{1 - (f_c / f)^2}$$

Problem 1 (40 pts)

A RHCP plane wave at 1.575 GHz from a GPS satellite is incident on the earth. The earth has $\epsilon_r = 4.0$ and is taken as lossless. The angle of incidence is $\theta_i = 45^\circ$.

Calculate the total power density in the reflected plane wave relative to that of the incident plane wave.



Room for work

Solution

The incident power divides equally between TE_z and TM_z parts. Therefore, if there is an incident power density of one $[W/m^2]$, then the incident power density in the TE_z and TM_z parts is 0.5 $[W/m^2]$. The power density in the reflected wave is therefore

$$P_r = \frac{1}{2} |\Gamma_{TM}|^2 + \frac{1}{2} |\Gamma_{TE}|^2.$$

From Snells' law, the transmitted angle is

$$\theta_t = 20.705^\circ.$$

We then have

$$Z_1^{TM} = \eta_0 (0.7071)$$

$$Z_1^{TE} = \eta_0 (1.4142)$$

$$Z_2^{TM} = \eta_0 (0.4677)$$

$$Z_2^{TE} = \eta_0 (0.5345).$$

This gives us

$$\Gamma_{TM} = -0.2038$$

$$\Gamma_{TE} = -0.4514.$$

We then have

$$P_r = 0.1227$$

Problem 2 (30 pts)

A coaxial cable has an inner radius of 1 [mm] and an outer radius of 5 [mm]. The inner wire is made of copper with a conductivity of 5.8×10^7 [S/m]. The outer conductor is a braided aluminum conductor that has an effective conductivity of 2.0×10^7 [S/m]. Both the copper and the aluminum are nonmagnetic. The filling material is Teflon which has a relative permittivity of 2.1 and a loss tangent of 0.001.

Calculate the attenuation in [dB/m] on the coaxial cable at a frequency of 1.0 [GHz].

Room for work

Solution

The propagation constant is given by

$$\gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)}$$

where

$$C = \frac{2\pi\epsilon_0\epsilon_r}{\ln\left(\frac{b}{a}\right)}$$

$$L = \frac{\mu_0}{2\pi} \ln\left(\frac{b}{a}\right)$$

$$G = \frac{2\pi\sigma_d}{\ln\left(\frac{b}{a}\right)}$$

$$R = \left(\frac{R_{sa}}{2\pi a} + \frac{R_{sb}}{2\pi b} \right)$$

and

$$R_{sa} = \frac{1}{\sigma_a \delta_a}$$

$$R_{sb} = \frac{1}{\sigma_b \delta_b}.$$

We then use

$$\text{Attenuation} = \alpha \left(\frac{20}{\ln(10)} \right) [\text{dB/m}].$$

The results are

$$C = 7.259 \cdot 10^{-11} \text{ [F/m]}$$

$$L = 3.219 \cdot 10^{-7} \text{ [H/m]}$$

$$G = 4.561 \cdot 10^{-4} \text{ [S/m]}$$

$$R = 1.76 \text{ } [\Omega/\text{m}]$$

This gives us

$$\gamma = 0.028403 + j(30.37168) \text{ [1/m]}.$$

From this we have

$$\text{Attenuation} = (0.028403) \left(\frac{20}{\ln(10)} \right) \text{ [dB/m]}.$$

This gives us

$$\text{Attenuation} = 0.246705 \text{ [dB/m]}.$$

Problem 3 (30 pts)

An air-filled rectangular waveguide has $b < a$. For a certain application, we want to make sure that all of the modes are attenuated sufficiently fast inside the waveguide, so that the entire field attenuates very fast. Note that all modes will attenuate at least as fast as the dominant mode, if it attenuates.

What is the largest value that the dimension a can be, so that the dominant mode is attenuated at a rate of at least A [dB/m] at a given frequency f ?

Your answer should be a formula that has the result for the dimension a in terms of A and f .

Room for work

Solution

In order to have the dominant more decay fast enough, we require

$$\text{Attenuation [dB/m]} = \alpha \left(\frac{20}{\ln(10)} \right) = A.$$

Hence we have

$$k_c \sqrt{1 - \left(\frac{f}{f_c} \right)^2} \left(\frac{20}{\ln(10)} \right) = A.$$

This gives us

$$\left(\frac{\pi}{a} \right) \sqrt{1 - \left(\frac{f}{\left(\frac{c}{2a} \right)} \right)^2} \left(\frac{20}{\ln(10)} \right) = A.$$

Squaring both sides, we have

$$\left(\frac{\pi}{a} \right)^2 \left(1 - \left(\frac{(2a)f}{c} \right)^2 \right) \left(\frac{20}{\ln(10)} \right)^2 = A^2.$$

Multiplying by a^2 , we have

$$\pi^2 \left(1 - \left(\frac{(2a)f}{c} \right)^2 \right) \left(\frac{20}{\ln(10)} \right)^2 = A^2 a^2.$$

Collecting terms, we have

$$a^2 \left(A^2 + \frac{4\pi^2 f^2}{c^2} \left(\frac{20}{\ln(10)} \right)^2 \right) = \pi^2 \left(\frac{20}{\ln(10)} \right)^2.$$

This gives us the result for the maximum allowed value of a , which is

$$a = \left(\frac{\frac{20\pi}{\ln(10)}}{\sqrt{A^2 + \frac{4\pi^2 f^2}{c^2} \left(\frac{20}{\ln(10)} \right)^2}} \right).$$

This may also be written as

$$a = \left(\frac{\pi}{\sqrt{A^2 \left(\frac{\ln(10)}{20} \right)^2 + \frac{4\pi^2 f^2}{c^2}}} \right).$$